

Hypothesis Testing for Two Sample Proportions

Statistics Worksheet · Grade 11–12 / Introductory College Statistics

Name: _____

Date: _____

Learning Objectives

- Calculate the pooled sample proportion ($\hat{p}_c$) for two independent samples
- Compute the z-test statistic for a two sample proportion hypothesis test
- Set up null and alternative hypotheses and draw conclusions for two sample proportion tests

Problems

1. In a study, Group 1 has 40 successes out of 200 trials and Group 2 has 60 successes out of 300 trials. Calculate the pooled sample proportion, $\hat{p}_c$.

$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2}$$

2. For Sample 1: 56 successes out of 2051, and Sample 2: 84 successes out of 2030. Calculate $\hat{p}_1$ and $\hat{p}_2$ (round to 4 decimal places).

$$\hat{p}_1 = \frac{x_1}{n_1}, \quad \hat{p}_2 = \frac{x_2}{n_2}$$

3. Using the Helsinki Heart Study data (Group 1: 56 heart attacks out of 2051 men on gemfibrozil; Group 2: 84 heart attacks out of 2030 men on placebo), calculate the pooled sample proportion $\hat{p}_c$. Round to 4 decimal places.

$$\hat{p}_c = \frac{56 + 84}{2051 + 2030}$$

4. A researcher wants to test whether the proportion of smokers who develop lung disease differs between two cities. Write the null hypothesis and the alternative hypothesis for this two-tailed test using p_1 for City A and p_2 for City B.

$$H_0 : p_1 = p_2 \quad \text{vs.} \quad H_a : p_1 \neq p_2$$



5. Using $\hat{p}_c = 0.0343$, $n_1 = 2051$, and $n_2 = 2030$, check all four normality conditions for the two sample proportion test. State whether each condition is satisfied (each product must be greater than 5).

$$n_1\hat{p}_c > 5, \quad n_1(1 - \hat{p}_c) > 5, \quad n_2\hat{p}_c > 5, \quad n_2(1 - \hat{p}_c) > 5$$

6. A quality control manager samples two production lines. Line 1 has 18 defective items out of 150 and Line 2 has 27 defective items out of 180. Calculate $\hat{p}_1$, $\hat{p}_2$, and $\hat{p}_c$.

$$\hat{p}_1 = \frac{18}{150}, \quad \hat{p}_2 = \frac{27}{180}, \quad \hat{p}_c = \frac{18 + 27}{150 + 180}$$

7. Using the quality control data from Problem 6 ($\hat{p}_1 = 0.12$, $\hat{p}_2 = 0.15$, $\hat{p}_c = 0.1364$, $n_1 = 150$, $n_2 = 180$), compute the z-test statistic for testing $H_0: p_1 = p_2$. Round to 2 decimal places.

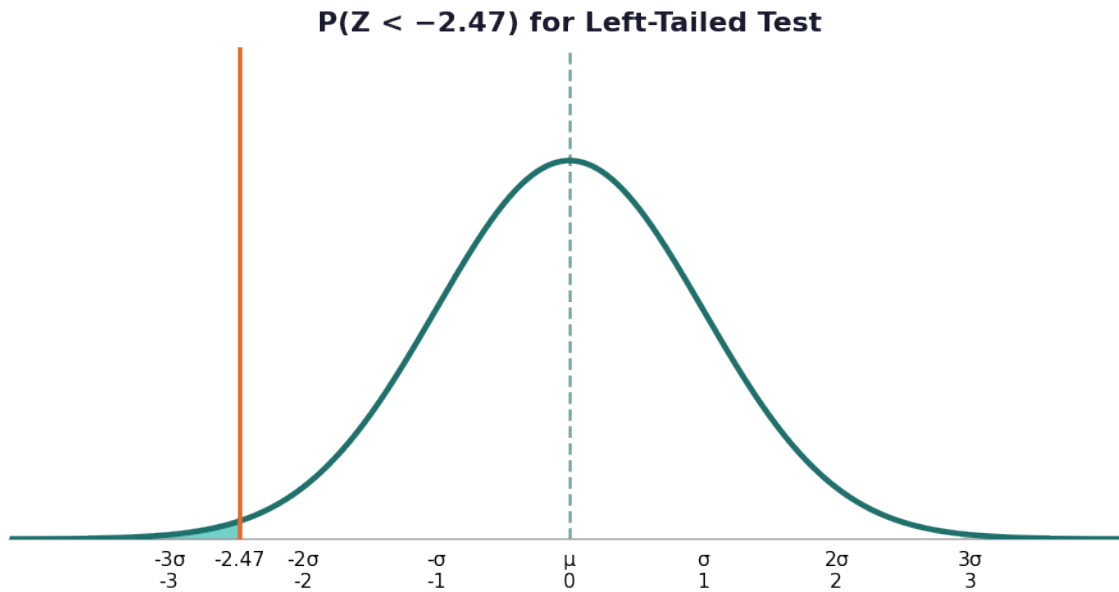
$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

8. Using the Helsinki Heart Study ($\hat{p}_1 = 0.0273$, $\hat{p}_2 = 0.0414$, $\hat{p}_c = 0.0343$, $n_1 = 2051$, $n_2 = 2030$), compute the z-test statistic. Round to 2 decimal places.

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

9. For the Helsinki Heart Study, the alternative hypothesis is $H_a: p_1 < p_2$ and the significance level is $\alpha = 0.05$. The computed z-test statistic is -2.47 . Using the standard normal distribution, find the p-value and state whether you reject or fail to reject the null hypothesis.





10. In an election poll, 480 out of 1200 voters in County A and 390 out of 900 voters in County B support a new policy. At a significance level of 0.05, test whether the proportion of supporters in County A is less than in County B. State the hypotheses, compute p-hat c and z, find the p-value (left-tailed), and state your conclusion.

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$



Hypothesis Testing for Two Sample Proportions — Answer Key

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Answer Key

1. Answer: $\hat{p}_c = 0.20$

- Identify values: $x_1 = 40$, $x_2 = 60$, $n_1 = 200$, $n_2 = 300$
 - $\hat{p}_c = (40 + 60) / (200 + 300) = 100 / 500 = 0.20$
-

2. Answer: $\hat{p}_1 \approx 0.0273$, $\hat{p}_2 \approx 0.0414$

- $\hat{p}_1 = 56 / 2051 \approx 0.0273$
 - $\hat{p}_2 = 84 / 2030 \approx 0.0414$
-

3. Answer: $\hat{p}_c \approx 0.0343$

- $\hat{p}_c = (56 + 84) / (2051 + 2030) = 140 / 4081$
 - $\hat{p}_c \approx 0.0343$
-

4. Answer: $H_0: p_1 = p_2$ (no difference); $H_a: p_1 \neq p_2$ (two-tailed)

- The null hypothesis always states no difference: $H_0: p_1 = p_2$
 - Since we are testing if proportions differ (no direction given), the alternative is two-tailed: $H_a: p_1 \neq p_2$
-

5. Answer: All four conditions satisfied; normality assumption is met.

- $2051 \times 0.0343 \approx 70.35 > 5 \checkmark$
 - $2051 \times (1 - 0.0343) \approx 1980.65 > 5 \checkmark$
 - $2030 \times 0.0343 \approx 69.63 > 5 \checkmark$
 - $2030 \times (1 - 0.0343) \approx 1960.37 > 5 \checkmark$
 - All conditions are satisfied; the normal approximation is valid.
-

6. Answer: $\hat{p}_1 = 0.12$, $\hat{p}_2 = 0.15$, $\hat{p}_c \approx 0.1364$

- $\hat{p}_1 = 18 / 150 = 0.12$
 - $\hat{p}_2 = 27 / 180 = 0.15$
 - $\hat{p}_c = (18 + 27) / (150 + 180) = 45 / 330 \approx 0.1364$
-

7. Answer: $z \approx -0.72$

- Numerator: $0.12 - 0.15 = -0.03$
 - $\hat{p}_c \times (1 - \hat{p}_c) = 0.1364 \times 0.8636 \approx 0.11779$
 - $1/n_1 + 1/n_2 = 1/150 + 1/180 \approx 0.00667 + 0.00556 = 0.01222$
 - Denominator: $\sqrt{0.11779 \times 0.01222} \approx \sqrt{0.001439} \approx 0.03794$
 - $z = -0.03 / 0.03794 \approx -0.79$
-

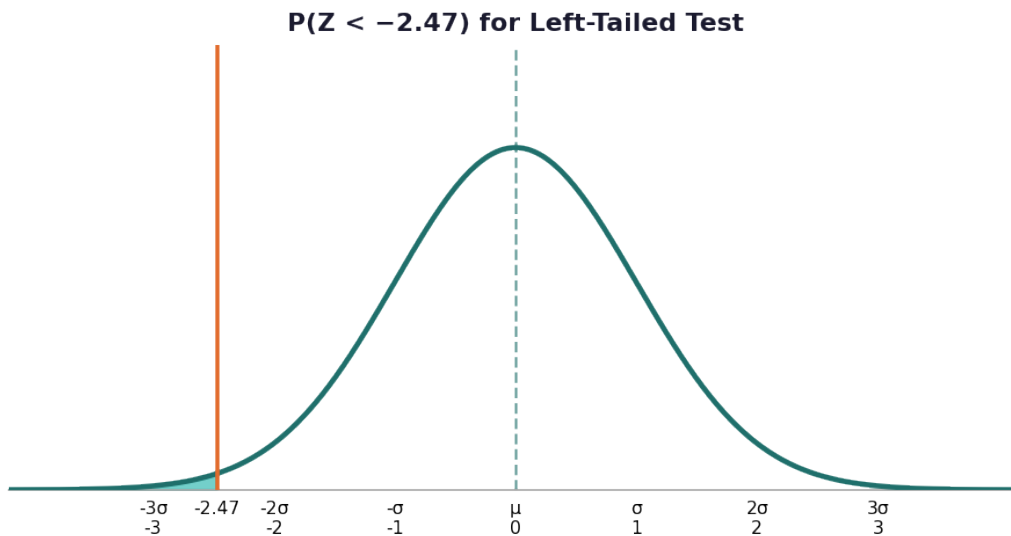
8. Answer: $z \approx -2.47$

- Numerator: $0.0273 - 0.0414 = -0.0141$
-



- $\hat{p}_c \times (1 - \hat{p}_c) = 0.0343 \times 0.9657 \approx 0.03312$
- $1/n_1 + 1/n_2 = 1/2051 + 1/2030 \approx 0.000488 + 0.000493 = 0.000981$
- Denominator: $\sqrt{0.03312 \times 0.000981} \approx \sqrt{0.00003249} \approx 0.005700$
- $z = -0.0141 / 0.005700 \approx -2.47$

9. Answer: p-value $\approx$ 0.0068; Reject H_0 at $\alpha = 0.05$



- This is a left-tailed test since $H_a: p_1 < p_2$
- $z = -2.47$
- From the standard normal table: $P(Z < -2.47) \approx 0.0068$
- Since p-value (0.0068) $< \alpha$ (0.05), we reject H_0
- Conclusion: There is sufficient evidence that the proportion of heart attacks in the gemfibrozil group is significantly less than in the placebo group.

10. Answer: $z \approx -1.47$; p-value $\approx$ 0.0708; Fail to reject H_0

- $H_0: p_1 = p_2$ vs $H_a: p_1 < p_2$ (left-tailed test)
- $\hat{p}_1 = 480/1200 = 0.40$; $\hat{p}_2 = 390/900 \approx 0.4333$
- $\hat{p}_c = (480 + 390) / (1200 + 900) = 870 / 2100 \approx 0.4143$
- $\hat{p}_c \times (1 - \hat{p}_c) = 0.4143 \times 0.5857 \approx 0.2427$
- $1/n_1 + 1/n_2 = 1/1200 + 1/900 \approx 0.000833 + 0.001111 = 0.001944$
- Denominator: $\sqrt{0.2427 \times 0.001944} \approx \sqrt{0.0004718} \approx 0.02172$
- $z = (0.40 - 0.4333) / 0.02172 \approx -0.0333 / 0.02172 \approx -1.53$
- $P(Z < -1.53) \approx 0.0630$; since $0.0630 > 0.05$, fail to reject H_0
- Conclusion: There is not sufficient evidence at the 0.05 level to conclude that County A has a lower proportion of supporters than County B.

